Exploring Data Sonification

Visualisation and Sensing BSC2 2025 | Tom Armitage | t.armitage@arts.ac.uk

In this exercise, you'll use a web-based tool to create sonification of data files. You'll learn:

- how time-series data can be mapped to simple sonification
- how data points can be mapped to parameters within a sonification
- how your own data could possibly be sonified

This tool provides a straightforward way into some of these ideas without necessarily writing code; all of them are highly achievable within the languages and platforms you've encountered, though.

Timing

This workshop is designed to take 45-60mins, including lecturer walkthrough.

Requirements

Web browser, headphones.

Getting Started

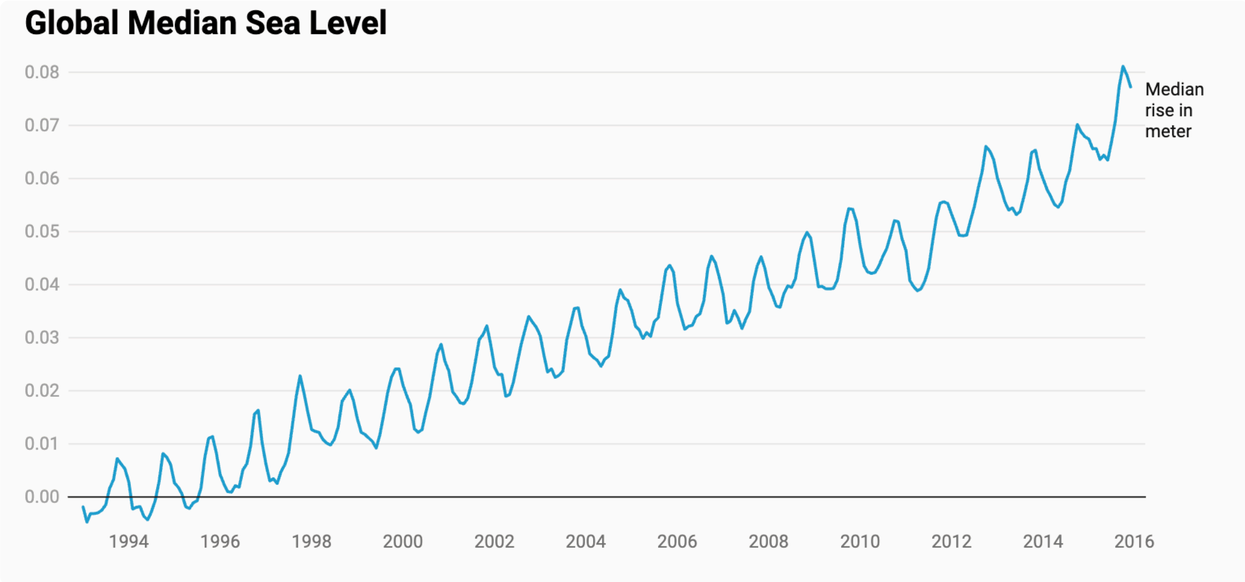
Download the zipfile of data from Moodle.

Go to https://studio.datasonifyer.de/en in a web browser; make sure your headphones are connected to your computer.

DataSonifyer allows you to import CSV files, and then map the contents of them to data.

When it loads, you will see some random data. You can push "play" top right to confirm your audio works.

1. Rising tide.



The European Space Agency have this data set of the *median rise in sea level in meters* by year.

You've got this dataset as `data-sealevel.csv`.

- Click "**Load Data**" and select this CSV file. For delimiter, just specify a comma (,) on its own.
- Select **"Membrane" as sound**. Set the **BPM to 220**. Turn loop **off**. Now play the sonification; the data points are played quickly one after the other.
- In each of the sound modules below, you can set whether the particular sound parameter is controlled by a field (column) in the data, or is "Custom" - which means it's set to an unchanging value.

Let's start adjusting some parameters. Click **add sound modules** at bottom, and tick the Frequency, Amplitude, Filter, Envelope and Effect boxes.

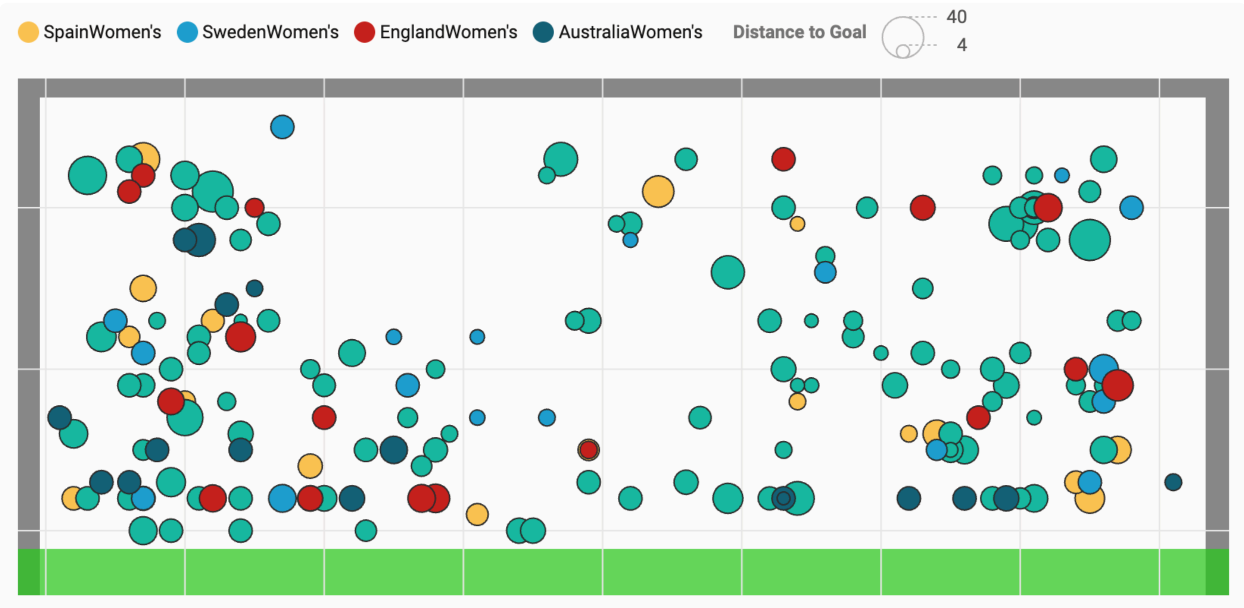
- **Frequency:** Set the data series to track "Median rise in Meter". Set the minimum audio frequency (Min Freq) to 110 and the maximum audio frequency (Max Freq) to 440. This is the distance between A2 and A4, i.e. two octaves. As no scale mode has been selected, the values will map to frequencies, not necessarily musical notes.
- **Envelope:** Select "Attack" as "Type" and set the "Custom" envelope to 1 second. This will create a slow swell asa it comes in. It'll also automatically make a slow release. If you listen to this, it'll probably boom and clip as all the sounds now overlap, so...
- **Amplitude:** turn the amplitude down! Set it to about 0.3 as a Custom parameter.
- **Filter:** Let's **also** control the filter based on sea level. Set "Median Rise in Meter" as "Data" and a "Lowpass Filter" as "Type". Set the filter cut-off frequency between 80 (Min Cut) and 1000 (Max Cut) - these are values in hz. This means that as the sea

level change goes up, the filter will open up, and the sound will get brighter.. This creates an effect as if the sound is gradually rising from the depths - like the level of the sea.

- **Effect:** Let's finally add some space to the sound with reverb. The stronger the reverb effect, the more reverberation you will hear. This can create a sense of distance. Select "Reverb" for "Type" and "Median Rise in Metres" for "Data". Then set Min FX to 0.9 and Max FX to 0.1. This builds an inverse relationship between the sea-level rise and reverb amount: smaller amounts of rise will have more reverb, and the larger amounts will have less reverb. As the sound gets higher, it'll sound closer, less distant, and like it's rising out of the sea.
- That's it! You can record your audio by clicking "record", then "play" at the top. When you are done, press stop, then click recording again; you can download your recording at the bottom of the page.
- Try listening to the sonification with with *noise* sound playing - this time, there won't be any pitch. You could layer these two sounds together in an audio editor like Audacity, to create something that sounds very like the sea - whilst also describing a dataset!

2. Goals

Here's a dataset of all the goals in the 2023 FIFA Women's World Cup. First, what it looks like visually:



In this screengrab, we can see the outline of the goal. The colour indicates the team; the position within the goal where the shot landed; and the size of the circle how far away the short was scored from.

The dataset is in your zipfile as `data-2023-womens-world-cup-goals.csv`.

In a new window, open DataSonifyer again, and load this data. As always, the separator is a comma.

Let's see what we can sonify here:

- we could represent the position of the goal left-to-right by **panning** the sound in the effects rack.
- we could represent the position of the goal bottom-to-top with **pitch**, lower to higher
- we could represent the distance the goal was scored from by the **duration** of the sound.

Add all six modules into your sketch. You should:

- set **frequency** to track **GoalHeight** - this time, use a musical scale. Set the octave limits to something that sounds good to your ears.
- use the **effects** module and the **panning** effect to map **GoalWidth** to panning.
- to set the duration, use an **Envelope** in **Duration** mode to track **DistToGoal** - be sure to set the range of min/max to something you feel is suitable.
- You'll want to adjust Amplitude to not make things sound bad.

The data is currently ordered by match, and then by minute - that is to say, each goal is just the next goal scored in the cup.

We could base the timing on something else to learn something new about the goals. We've got the **minute** of the match each goal was scored.

- Add the Rhythm module, and set its Data to track **minute**.
- You might want to speed up the track, set it to 180bpm or so.

Now listen to the sonification. This imagines all the goals being scored in the same match. Towards the end, can you hear anything unusual? Do you have any idea what that might be?

3. Stretch Goal: Your Own Data

If you have a CSV to hand of your own data from your project, explore sonifying it with these tools. You might want to make a **simpler** CSV file, using a spreadsheet tool, pandas, or similar, just to get it down to a few columns you'd like to explore.

Be sure to record anything interesting that emerges!